Introduction

> Undoubtedly his greatest contributions were in the field of psychohistory. Seldon found the field little more than a set of vague axioms; he left it a profound statistical science.
>
> Isaac Asimov, *Foundation* (1942)

Danger is implicit even in the metaphor. *Mining* refers specifically to the extraction of valuable metallic ores from the earth, and metaphorically to any process that extracts the rare and valuable content from its surrounding context, usually at the cost of extreme expenditures of labor and at danger to the environment. *Data mining,* a similarly intensive operation with implicit hidden costs, also begins with counting, but includes statistical transformations to test for relationships such as correlation and significance. *Text mining,* a process with all the dangers of the former two categories, mines for value with text as the data. It begins with computational transformations that break up and classify digitalized strings of archival text into units representing constituent words and phrases, and then applies statistical manipulations that enable us to study the kinds of meaningful signals and relationships that trained analysts from the humanities and social sciences would like to detect in the course of reading. With the help of computers, the insights of reading can be carried out over a larger scale.

Text mining is powerful if done right. When rendered as visualizations, the output of text mining offers, at least potentially, an outsized return on investment, distilling shelf miles of text into a valuable, pithy representation of what all those words *meant.* A single visualization might reduce the story of how a single institution's politics have changed over a hundred years. Scholars from the humanities have used text mining to give us a mirror of what stories people have told each other about how Covid is transmitted, and how American novelists present white characters and black characters differently over time.[1]

[1] Shadi Shahsavari, Pavan Holur Timothy R. Tangherlini, and Vwani Roychowdhury, "Conspiracy in the Time of Corona: Automatic Detection of Covid-19 Conspiracy Theories in Social Media and

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We see an example of this method in Figure 0.1, where word embeddings have been used to show how members of the US Congress talked about the term "slavery" from 1870 to 2010. The *x*-axis is a timeline from 1870 to 2020 (the full length of the dataset), divided into five-year periods. The *y*-axis gives an abstract measure of "similarity" between the keyword "slavery" and words that appeared in the same sentence, paragraph, or debate. Words higher on the *y*-axis appeared alongside "slavery" with greater regularity. The diagram was designed to show the shifting context in which the word "slavery" appeared from one generation to the next.

The big takeaway of Figure 0.1 is that for most of the period since the Civil War, Congress has not talked about race when it has spoken about slavery. Members of Congress used the term "slavery" as a metaphor for a host of other ills they wished to denounce. They classified Mormon polygamy, Naziism, and Bolshevism forms of mental or moral enslavement.

But much less frequently did members of Congress acknowledge that slavery as an institution had wrecked the lives of generations of African-Americans, brought to the United States against their will. Nor did members of Congress talk about the legacy of slavery in modern America. Instead, they associated slavery with developments abroad, especially the "economic slavery" that they associated with communist regimes.

Congress's reluctance to talk about the facts of American history in the chief organ of American democracy persisted for over a century. Only in the 1990s did "racism" and "lynching" become collocates of "slavery" for the first time.

A diagram like Figure 0.1 is designed to dramatize shifts over time, shifts that may be so subtle that they surprise many an American historian.[2] It is not the algorithm alone that renders such an insight obvious. The algorithm has to be deployed by a coder who understands the significance of change over time to patterns of words that appear with other words. Unlocking the mysteries of the algorithm, the category of time reveals how we have lied to each other about our institutions and identity.

There are also many data-driven analyses of text that purport to offer substantive insight – and fail. Such failures of text mining occur when

the News," *ArXiv Preprint ArXiv:2004.13783*, 2020; Richard Jean So, *Redlining Culture* (New York: Columbia University Press, 2020);

[2] I am not myself an American historian, but I record the relative surprise of my collaborator, Peter Kastor, with whom I am writing about word embeddings in Congress.

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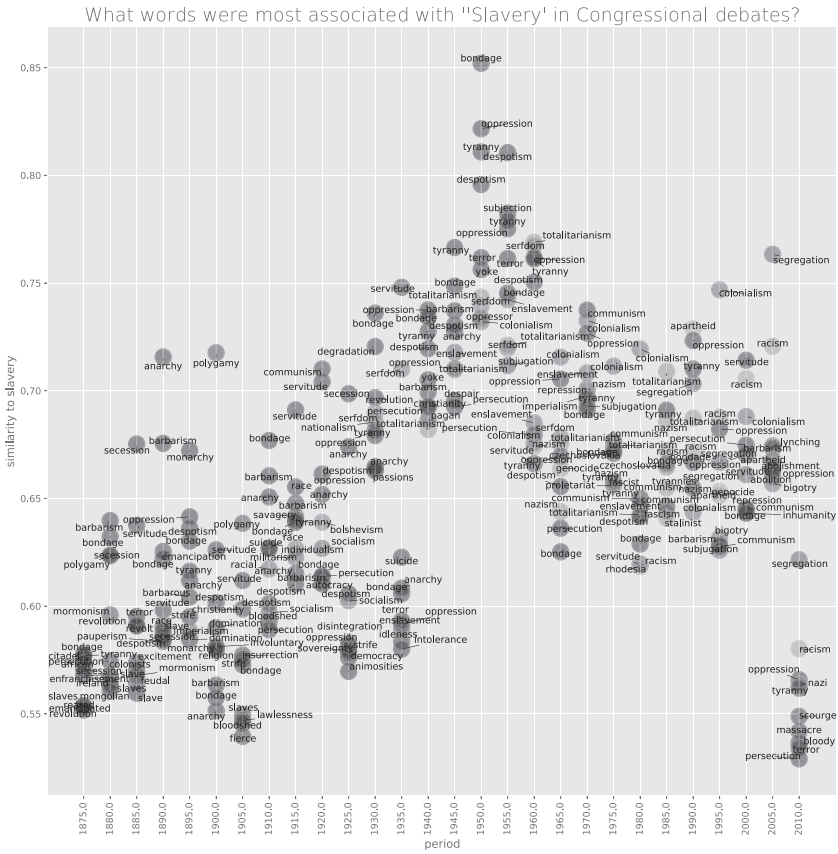


Figure 0.1 Words co-located with the keyword "slavery" in the debates of the US Congress. *X*-axis indicates time; *y*-axis indicates an abstract measurement of "similarity" measuring how consistently the words are invoked together. Detected using Gensim word embedding software.

algorithmic distillations of text are misapplied, with the result of analyses that are empty, biased, or simply false.

Consider the fate of the wordcloud in recent years. The wordcloud (sometimes known as a "wordle" before the arrival of a popular game of the same name) is a visualization where the size of words corresponds to their frequency. In the wordcloud visualization in Figure 0.2, the larger words are at the center, but the words' spatial arrangement is random, a stylistic choice that encourages the viewer's eyes to wander the cloud, free associating along the way.

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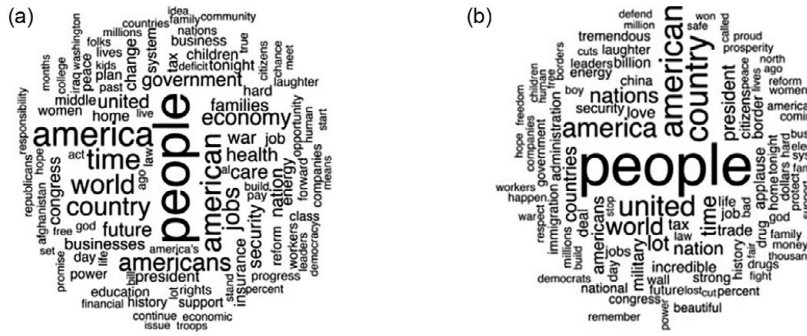


Figure 0.2 Data-driven visualizations that obscure rather than illuminate meaningful differences: Wordclouds for the official speeches of a) Barack Obama and b) Donald Trump during their presidencies. Source: "Presidential Speeches" dataset, https://millercenter.org/the-presidency/presidential-speeches. Visualized by author.

After first appearing as a navigation tool on the photo-sharing website Flickr in 2004, wordclouds became a preferred visualization used by newspapers for reducing political speeches into a pop graphic. From 2004 to 2017, the wordcloud was everywhere. Bloggers and websites used wordclouds as a navigation index. During the Obama presidency, newspapers such as *The Guardian* regularly ran visualizations distilling Barack Obama's State of the Union speeches into a wordcloud. But the wordcloud abruptly disappeared from newspapers and blogs alike when political events tested the limits of the tool to reveal the truth.

As late as the Obama presidency, the wordcloud served the public as an easy tool for rendering complex text into a simple snapshot of a historical moment. In Obama's speeches, keywords such as "freedom," "people," and "journey" loomed large in size, reinforcing Obama's sense of America as a nation at a turning-point in history.[3] Just as importantly, the wordcloud promised to distill into a single visualizable truth the reality of truth in an era of information overload.

Soon after Donald Trump's election, however, the fad for wordclouds began to dissipate. Under a new president, language was being used in a fundamentally new way, but wordclouds failed to do justice to the difference. Indeed, had journalists in 2017 the power to compare all of Trump's speeches from his presidency with Obama's, they would have discovered a

[3] "Inauguration Speeches in Wordclouds," *The Guardian*, January 21, 2013, sec. US News, www.theguardian.com/world/gallery/2009/jan/21/obama-inauguration-word-clouds-gallery.

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mysterious fact: the two men used *very many* of the same words at the same rate. Both proclaimed their allegiance to the "American" "people" and their "country;" both referenced the "United" "States." Obama talked about the "economy" and "jobs," while Trump talked about "trade" and "jobs." They both referenced "citizens," "families," and "children." Both touted their commitment to "security," to America's role in the "world," and to the importance of the "time" of the present.

The more Trump spoke, the clearer the weakness of the wordcloud became. The word cloud, a tool for showing the most frequent words, did not in its automatic form look for phrases. It could not tell the difference between Obama's "right here in America" – where he intended to create jobs – and Trump's "make America great again," a plea that barely veiled a longing for a time of white supremacy.

In the early years of the Trump presidency, journalists and politicians kept running Trump's speeches and tweets through the visualization, hoping for different results. When that approach failed, one democratic representative in Congress tried a new strategy to capture Trump's abrasive style, creating his own wordcloud generated from a list of the individuals and organizations that Trump had attacked over Twitter.[4] But mainly the wordclouds of 2017–21 continued to backfire. One published wordcloud distilled the keyword "Klansmen" as a word related to Biden, but not related to Trump. In the accompanying article, a journalist reassured his readers that future President Joe Biden had condemned the seeming condoning of the KKK by Donald Trump, who himself never mentioned the Klan by name.[5] By then, it must have become clear: Trump had broken the wordcloud.

By aping his opponents' populist language for nationalist and racial purposes, Trump had produced a close parody of Obama's invocations of patriotism. The result was that Trump's speech was virtually indistinguishable from Obama's – at least in wordcloud form. The wordcloud betrayed how Trump's broadly inclusive language appealed to many American's sense of patriotism, indeed, wooing some voters who had cleaved to Obama, even while carefully encoded dog-whistles summoned hate groups

[4] Jessica Estepa, "House Democrat Makes a Word Cloud of Trump Insults," *USA Today*, September 26, 2017, www.usatoday.com/story/news/politics/onpolitics/2017/09/26/house-democrat-makes-word-cloud-trump-insults/705120001.

[5] Ben Schott, "Biden and Trump Go Cloud to Cloud," *Bloomberg*, August 28, 2020, www.bloomberg.com/opinion/articles/2020-08-28/biden-and-trump-nomination-speeches-are-two-clashing-word-clouds.

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to his side. For anyone paying attention, the wordcloud might have been trying to tell us something about the political malleability of populist language. For the purpose of explaining to newspaper readers the differences between two presidents, however, the wordcloud was kaput.

Understanding the priorities, values, and falsehoods of president is a matter of no small priority. Reasonable people may reasonably disagree on political values. But newspapers that serve reasonable people try to provide everyone with the same basic facts about the present moment and recent past. The wordcloud vanished because it proved incapable, at least in its automatic form, of generating straightforward facts about Trump that were capable of picking up on the president's many obvious differences from his predecessor.

A world trained on the manipulation of data but ignorant of, or insensitive to, the data's original context will be dangerously cut off from reality. Text mining can aid us by helping humans become more accurate at sifting data at scale. Without the insights of the humanities, text mining falls readily into error: it is a monkey reaching for the moon, guided only by the moon's reflection in a pond, where he will never grasp it.

That we should be interested in accuracy is obvious. Universities have long aspired to be the grounds where children reared in the native bias of their own locale learn to form an acquaintance with a broader reality. Companies that depend on the parsing of language with code should likewise be interested in the forms of hybrid thinking – humanities and mathematics, social science and informatics.

Text mining can be used to paint a portrait of partisan activity in legislative debates, newspapers, and social media sites. It can also support the larger method of using data to locate a discontinuity in time and of learning as much as possible about the distinctiveness of the event relative to other similar disruptions, as well as the individuals, causes, and contexts of disruption.

Once we've turned data to the purpose of understanding how two moments in time differ, we've taken a great step towards reality. We have fit data to the purpose of asking a real historical question. We've provided an instrument that is practically useful for assigning blame and pursuing reform and, even better, for starting to conceptualize the changes that determined the shape of the present.

Why were solutions of this kind invisible to the staff of American newspapers during the Trump years, when the limitations of the wordcloud began to show? They were invisible because text mining for historical analysis was still in its infancy during the Trump years. The

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place where the new tools would appear was still a no man's land between two enemy encampments.

The No Man's Land Where Better Tools Come From

In the years of the Trump Administration, a vast canyon extended between the experts in quantitative knowledge and that collection of scholars in the liberal arts who study documents for tiny signals of cultural change over time. Let us call these individuals "The Temporal Cartographers' Society." It is they who would face no difficulty parsing the differences between Obama's speeches and Trump's. The Society's ranks include professional historians, literary scholars, and historical sociologists, as well as some of the most talented members of the world of journalism and law. Mainly, they trained in liberal arts subjects, where, if quantitative methods are used to analyze text, they have been introduced (or reintroduced) only over the last two decades.

Those fields have long traditions of struggle and debate devoted to the difficulty of accurately summarizing a thousand letters. Historians' standard of truth, for instance, requires them to carefully reflect on the biases and silences of the documents with which they work, to worry over the circumlocutions and hidden suggestions of text. They also worry over hidden dimensions of change to the historical context in which words are invoked, a problem not easily reduced to the words on the page. As the scholar William Sewell once explained, not only words and ideas, but also new identities structure society over time. The result is the laws of change may work differently in one generation than another.[6] For this reason, historians study the imperfections, distortions, and omissions that riddle records from the past – including circumlocutions like Trump's phrase "make America great again," a phrase that must be unpacked beyond its surface invocation of prosperity for the implied longing for an era of white supremacy and male dominance to surface. The Temporal Cartographer's Society, in other words, has a lot to offer any study that depends on absolute honesty about what we know, what we don't know, and where data has been distorted by the bias of the past.

Traditionally, members of the Temporal Cartographers' Society have lacked tools for generalizing about billions or trillions of words, how those words cluster and change. They have adopted admirably to other forms of work with data – to birth and death rates, or what quantities of nutrition it was possible to buy on a working man's salary, or counting pollen samples

[6] Sewell, "Three Temporalities," 245–80.

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to chart changing temperatures.[7] Even with data, however, this group tends to be hesitant about over-grand generalizations. They rarely look for laws of historical truth. When they turn to numbers, they examine moments of change with enormous precision.

Generalizations at broad scale are the domain of another group, namely the engineers, mathematicians, statisticians, and computer scientists whose conceptual domain we might call the "world without history," because thinking about cultural and political revolution represents so little of the work done in that domain.

The world without history is the imaginary territory that is home to students of statistics and machine learning. Their labors support the design of computers and the creation of a thousand tools for animating design, medicine, and neuroscience. And some of its scions apply math on the stock market, where they sometimes leverage historical data while promising to predict the future.

The world without history has its own blindspots. In my experience, few computer scientists are exposed to historicist reasoning anywhere in their training. They often lack the broad literacy in the humanities one needs to understand the potential and the limitations of these conceptual models. And this ignorance of computationalists has made trade – let alone conversation – between the world without history and the Temporal Cartographers' Society virtually nonexistent for many years.

The Temporal Cartographers' Society entertains a certain skepticism towards the world without history. Issues of race, gender, the silences of the archive, and the costs of labor highlight the exploitative implications latent in the phrase "text mining," which refers to the process of computationally extracting meaning from digitalized documents. As Kate Crawford has pointed out, the metaphor of "mining" underscores the promise that computational labor will yield precious resources and therefore merits corporate investments.[8] But as historians know, mining industries have also been classical sites of the exploitation of indigenous, ethnic, and other working-class labor, entailing the making invisible of laboring bodies often broken, suffocated, and poisoned by working conditions. Mining industries have also been sites of lasting environmental pollution,

[7] Steven Ruggles, "The Revival of Quantification: Reflections on Old New Histories," *Social Science History* 45:1 (October 1, 2021): 1–25, https://doi.org/10.1017/ssh.2020.44; Sam White, Christian Pfister, and Franz Mauelshagen, *The Palgrave Handbook of Climate History* (London: Palgrave, 2018).

[8] Kate Crawford, *The Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence* (New Haven, CT: Yale University Press, 2021).

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as well as the origin point of the fossil economies, dependence on which has produced the carbon emissions of recent centuries, resulting in the climate emergency. To insist on referring to the work of computational history with texts as "text mining" is to testify to the fact that computational enterprises are implicated in potentially exploitative relationships with labor and the environment.

There is a longer story behind the distance between the two worlds as well. In American history departments, many faculty regard with distaste previous encounters with the methods of the world without history. The so-called quantitative turn of the 1960s and 1970s is remembered today as a moment when millions of dollars supported vast research collaborations in history and adjacent social science fields. The Philadelphia Project, a study of social mobility and the American Dream, is remembered chiefly as an expensive disaster whose authors failed to publish meaningful work. More notorious still is Robert Fogel and Stanley Engerman's *Time on the Cross*, which used a data base of life-expectancy to argue that enslaved Americans were treated better than their working-class counterparts in the north.[9] The numbers in Fogel and Engerman's project included 0.7 as the number of times that the average enslaved person was whipped per year at a single plantation in Louisiana between 1840 and 1842.[10] The crudely racist reductionism associated with these projects – which offered a decimal in the spot that could be occupied by tomes on the separation of families of enslaved people, the experience of the body in pain, and intergenerational trauma – still circulates in history departments today as evidence of what can go wrong when the desire for quantitative accuracy overrides humanists' appreciation of the human condition. Even as nearby fields such as political science and sociology have embraced big data, with scholars refining the methods that illuminate the major theories of the field, the refinement of similar tools for dealing with big data in history has only just begun.[11]

[9] For the Philadelphia Project, see Luke Blaxill, *The War of Words: The Language of British Elections, 1880–1914*, Royal Historical Society, Studies in History. New Series 103 (Woodbridge; Rochester, NY: The Boydell Press, 2020); Robert Fogel and Stanley Engerman, *Time on the Cross* (Boston, MA:Little, Brown and Company, 1974).

[10] Blaxill, *The War of Words*, 22 n. 9.

[11] For the comparative evolution of political science and history and broader meditation on quantitative methods suited to data, see Gregory Wawro and Ira Katznelson, *Time Counts: Quantitative Analysis for Historical Social Science* (Princeton, NJ: Princeton University Press, 2022). I draw on innovative work from political science and sociology throughout this book, but two books stand out as recent statements about methods in the field: Brandon Stewart, Justin Grimmer, and Margaret E. Roberts, *Text as Data* (Princeton, NJ: Princeton University Press, 2022), and Xiaoling Shu, *Knowledge Discovery in the Social Sciences: A Data Mining Approach* (Berkeley, CA: University of California, 2020).

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As recently as the Trump years, historians' warnings about the dangers of quantitative tools and information scientists' patterns of selective study meant that a dangerous and underexplored territory sprawled between the world without history and the domain of the Temporal Cartographers' Society. Few travelers crossed from one domain to the next. Much rarer was a compelling paper embraced by both worlds as the signal of a new field ripe for investigation.

Today, promising things are afoot in this no man's land. Modern "digital historians" who use text mining in their work, for instance Luke Blaxill, readily credit the linguistic turn as the inspiration for their own careful use of wordcount and statistics to discern moments of change in political speeches. The space of "digital history" that applies computational tools to historical problems only now has its first journals – *Current Research in Digital History* was founded in 2018 and the *Journal of Digital History* was founded in 2021. Luke Blaxill's 2020 book, *The War of Words*, is the first historical monograph to use text mining as its major method.[12]

The truth is that scholars in the liberal arts are continuing to profit from and learn from debates about quantitative reasoning in the past, and they have many insights to offer those who work with data. African-American and other postcolonial scholars – especially Jessica Marie Johnson and Roopika Risam – have taken a leading role in describing the possible abuses of quantified perspectives on the experiences of enslaved or otherwise silenced and exploited communities in the past.[13]

Distant reading and the promise of digitalized data have attracted the attention of historians since at least Lincoln Mullen and Keller Funk's "Spine of the American Law" (2016), where legal historians adopted algorithms to detect important turning-points in the history of American case law. Since that point, the range of algorithmic tools for textual analysis available to digital historians has expanded enormously, and a thriving community of digital historians have begun to investigate the best practices of text mining.[14] Researchers in History have begun to apply artificial

[12] Blaxill, *The War of Words*.

[13] Jessica Marie Johnson, "Markup Bodies: Black [Life] Studies and Slavery [Death] Studies at the Digital Crossroads," *Social Text* 36:4(137) (December 1, 2018): 57–79, https://doi.org/10.1215/01642472-7145658; Roopika Risam, *New Digital Worlds: Postcolonial Digital Humanities in Theory, Praxis, and Pedagogy* (Evanston, IL: Northwestern University Press, 2019), www.jstor.org/stable/10.2307/j.ctv7tq4hg.

[14] The field of digital history has only just emerged as a proper field, with its first monograph in text mining (Blaxill, *The War of Words*) and the first monograph on using GIS applied to a historical question (Cameron Blevins, *Paper Trails: The U.S. Post and the Making of the American West* (New York: Oxford University Press, 2021).

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intelligence (AI) tools to the history field, and my publications have explored a range of different algorithmic approaches. Many of my own technical articles demonstrate positive proof of the effectiveness of natural language processing (NLP) and statistical measures for discovering hitherto unknown historical events in the history of capitalism.[15]

The research is truly "hybrid" in the sense of representing a new undertaking that fits neither into traditional categories of history nor of information science. This research requires scholars to formulate new methods for validating whether and how each AI tool can support the high standard of reasoning required by the history profession. [16] This body of work, as a whole, points the way towards new standards of believability as well as new opportunities for building and measuring historically-significant moments whose traces are embodied in textual data.

The expertise of the new hybrid field is sometimes called the "digital humanities" or "the digital social sciences"; and one corner of it – trained on pattern-recognition in images, movement, maps, video, audio, and text alike – is sometimes described as "cultural analytics."[17] The expertise in text mining

[15] Jo Guldi, "Parliament's Debates about Infrastructure: An Exercise in Using Dynamic Topic Models to Synthesize Historical Change," *Technology and Culture* 60 1 (March 21, 2019): 1–33; Jo Guldi, "The Official Mind's View of Empire, in Miniature: Quantifying World Geography in Hansard's Parliamentary Debates," *Journal of World History* 32:2 (2021): 345–70, https://doi.org/10.1353/jwh.2021.0028; Jo Guldi, "The Algorithm: Mapping Long-Term Trends and Short-Term Change at Multiple Scales of Time," *The American Historical Review* 127:2 (June 1, 2022): 895–911, https://doi.org/10.1093/ahr/rhac160.

[16] Dzovinar Kévonian and Philippe Rygiel, "Connected Ogres: Global Sources in the Digital Era," *Monde(s)* 21:1 (May 11, 2022): 73–79; Jo Guldi, "The Common Landscape of Digital History," *Digital Histories: Emergent Approaches within the New Digital History* (Helsinki: Helsinki University Press, 2020); Ashley S. Lee, Poom Chiarawongse, Jo Guldi, and Andras Zsom, "The Role of Critical Thinking in Humanities Infrastructure: The Pipeline Concept with a Study of HaToRI (Hansard Topic Relevance Identifier)," *Digital Humanities Quarterly* 014:3 (2020). Jo Guldi, "The Measures of Modernity. Word Counts, Text Mining and the Promise and Limits of Present Tools as Indices of Historical Change," *International Journal for History, Culture and Modernity* 7 (November 3, 2019), https://doi.org/10.18352/hcm.589; Jo Guldi, "The Modern Paradigms of Explanation: Significance, Agency, and Writing History in the Era of Climate Change," *Isis* 110:2 (2019): 346–53; Ashley Lee, Jo Guldi, and Andras Zsom, "Measuring Similarity: Computationally Reproducing the Scholar's Interests," *ArXiv:1812.05984 [Cs]*, December 14, 2018, http://arxiv.org/abs/1812.05984; Jo Guldi and Benjamin Williams, "Synthesis and Large-Scale Textual Corpora: A Nested Topic Model of Britain's Debates over Landed Property in the Nineteenth Century," *Current Research in Digital History* 1 (2018), https://doi.org/10.31835/crdh.2018.01; For validation, Jo Guldi, "Critical Search: A Procedure for Guided Reading in Large-Scale Textual Corpora," *Journal of Cultural Analytics* (2018): 1–35, https://doi.org/10.22148/16.030.

[17] The wider field of the digital humanities includes practices such as metadata creation, mapping, network analysis, and the creation of online experiences and games. Cultural analytics focuses on the analysis of data – typically textual, musical, or visual – to produce insights about humanities questions, for instance the nature of cultural bias or the reality of change over time. For the digital humanities more broadly, see Eileen Gardiner and Ronald G. Musto, *The Digital Humanities: A Primer for Students and Scholars* (Cambridge University Press, 2015); for cultural analytics, see

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in particular commands a certain corner of cultural analytics, and the problem of historical truth adds another still more specific dimension of concern.

But history is not of concern only to historians. It is also a method embraced across the humanities and social sciences, relevant to law schools and schools of journalism, and useful in practically any endeavor where the investigation of human culture and institutions is at stake. I prefer to refer to the activities covered by this book as "text mining for historical analysis." "Text mining" is a province of research in AI, and there are researchers in computer science and informatics departments dedicated to the question of what makes text mining accurate and robust. Its practices borrow richly from the interdisciplinary cultural analytics community as well as practitioners of text mining who work across the social sciences, especially in political science and sociology. "Historical analysis" marks out diachronic text mining, trained on the modelling of temporal experience in all its variation, from styles of text mining pursued for purposes wherein time does not necessarily factor, for instance modeling fundamental differences between different languages and linguistic groups. Insights derived from historical analysis are relevant to scholars from history, art history, literature, political science, and economics, as well as practitioners at the law and business schools. Robust historical analysis is almost always improved by the insights of history scholars. The phrase "text mining for historical analysis" thus invokes a wide zone where text mining is used for pragmatic reasons. Unlike the phrase "digital history," text mining for historical

Lev Manovich, "The Science of Culture? Social Computing, Digital Humanities and Cultural Analytics," *CA: Journal of Cultural Analytics*, May 23, 2016, http://culturalanalytics.org/2016/05/the-science-of-culture-social-computing-digital-humanities-and-cultural-analytics. For a general introduction to the digital humanities, see Johanna Drucker, *The Digital Humanities Coursebook* (London: Routledge, 2021), and the various editions of Lauren Klein and Matthew Gold, *Debates in the Digital Humanities* (Minnesota, MN: University of Minnesota Press, 2004–20). For the long tradition of quantitative measures being used for literary criticism in the past, see Brad Pasanek, "Extreme Reading: Josephine Miles and the Scale of the Pre-Digital Digital Humanities," *ELH* 86: 2 (2019): 369, 375. The digital social sciences, by contrast, have tended to come from departments of sociology and political science and they are experiencing their own revolution in text mining applied to traditional questions motivated by social theory. See Peter Bearman, "Big Data and Historical Social Science," *Big Data & Society* 2:2 (December 27, 2015), https://doi.org/10.1177/2053951715612497; James A. Evans and Pedro Aceves, "Machine Translation: Mining Text for Social Theory," *Annual Review of Sociology* 42:1 (2016): 21–50. The discipline of history has traditionally claimed membership both in the humanities and the social sciences, and typically draws on the methods of both parts of the university. For a general introduction to the variety of approaches used in digital history, including maps, images, and text from the past, see Hannu Salmi, *What Is Digital History?* (London: Polity, 2020). In this book I will invoke the word "humanities" broadly for interpretive issues in contrast to quantitative ones, even though many of the approaches I foreground are of interest to the social sciences or were developed within social science frameworks.

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analysis carries no suggestion that computer-driven history will somehow substitute for history by traditional means – a threat that some members of the Temporal Cartographers' Society are all too wary of. Text mining for historical analysis rather sits across the many domains that use text, AI and statistics to produce certain kinds of knowledge, useful to certain purposes.

The hybrid thinking that has emerged in between quantitative and qualitative disciplines combines insights from history with those from information science has the power to catalyze a revolution in smart data. It is *hybrid* rather than merely interdisciplinary because what is produced in the process is more than an adjustment to old conversations. Something genuinely new is being produced where old fields meet, and this hybrid practice has the power to produce insight about large-scale troves of texts, not unlike the hybrid ears of corn that revolutionized industrial agriculture in the middle of the twentieth century.[18] Such a practice as this has a great deal to offer the community of scholars, although it is hard to see text mining replacing the many forms of work with data, documents, and archives that mark the specialty of history. The greatest beneficiaries of a hybrid practice of historical thinking with data are outside the walls of the academy.

Toward a Smarter Data Science

Data science today needs to become smarter in ways that benefit from learning that originates outside the study of data.

In Chapter 1, I tell the story of how my students plotted the trend of mentions of "ignorant women" over the course of debates from Britain's parliament in the nineteenth century. They concluded that women had become more and more ignorant over the course of the century – rather than reasoning as a historian, that Britain's parliament had become more and more biased against women over the course of the same time.

I learned about the dangers of text mining because I welcome students of engineering, statistics, and business, as well as history and literature majors, to my classroom. The gap in the education of a significant

[18] As science and technology scholars will note, "hybridization" is, like "mining," a deliberately ambivalent metaphor. Hybrid grains and the industrialized farms they supported drove smaller farmers from business in many parts of the world, helping to produce a global housing crisis and making agriculture overall more vulnerable – even if it was also more productive at scale. A monocrop of hybrid digital history would result in similar imbalances in the name of scale. There is a great deal more to be said that will not be said here. My intention in this volume is to show that hybrid practice can be done in a *more* responsible way, but I also underscore the need for the perseverance of traditional history and the teaching and practice of traditional methods.

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proportion of my students is a warning sign. If some of my cleverest students are making those mistakes today, then the titans of industry will be making similar mistakes tomorrow.

The solution to bad graphs and foolish interpretations is, I believe, a return to reality. Reality, in terms of our collective experience with each other, is perhaps only known through stories about our past, and there are many other kinds of stories that only speak to desire, its reflections, and shadows.

This is why hybrid knowledge matters. It's why any university that's teaching data science urgently needs to plan programs of instruction that foreground not only the skills of working with data at speed but also understanding what data means. We *can* use data to discover important trends, recent and older, that give us insights to the subtle changes that are shifting society. But to use those tools effectively, we need smart data and hybrid knowledge, attuned to the latent bias of the data and of algorithm and how to navigate them. Data-driven analysis of language, without a historical method, is apt to produce unusable information. Without help from abroad, the world without history is in trouble.

In spaces like digital history where computationalists and historians meet and ask questions about each other's basic premises, problems tend to emerge that were previously unknown to either field beforehand. When historians consult with algorithms, for example, they tend to move deliberately, reflecting on the fit of each algorithm for each separate stage of analysis. They care about distinguishing the signals of correlation from those of causation. These conversations can be productive of new insight and opportunities for rigor. A truly hybrid research process forces colleagues to submit their unstated hypotheses to careful testing. It requires refraining from hasty diagnoses and cheap answers until the process for arriving at truth meets the standards of fields that have never previously reviewed their terms of agreement.

The birth of new fields where old disciplines meet is a reality of our time. Data science is sometimes presented, in its ideal form, as the application of statistics to big data with substantive knowledge.[19] For this reason, teams of medical researchers work alongside statisticians and memory specialists to produce the field of biostatistics; teams of linguists collaborate with computer scientists and statisticians to produce computational linguistics; and political scientists and social scientists work with

[19] Ted Underwood reviews this characterization, which he credits to Drew Conway, CEO of Alluvium, in a blog entry: https://tedunderwood.com/2017/01/29/two-syllabi-digital-humanities-and-data-science-in-the-humanities.

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technology experts to produce computational social science. Now, it seems clear, a new field is emerging that engages text mining with the aim of establishing historical truth from the humanities and history in particular.

Hybrid thinking about historical truth has much to offer the rest of the world. It can aid political journalists who are constantly seeking new ways of understanding cultural differences in the electorate. It can support media streaming companies that wish to learn more about who identifies with certain kinds of music, lyrics, and narratives, and what kinds of content leaves them cold. These practices can entertain and educate the general public, which is fascinated by data visualizations that show them aspects of their own experience from unexpected angles. And it can support scholars already dedicated to a careful and critical analysis of *who* human beings are, *what* was changing at any given movement in the past, and *where* culture appears to be heading.

Yet most of the tools from data science today have been developed without consultation with ideas about truth from history or the humanities. Like the wordcloud, they are simplistic approximations of truth where information is lost. Mathematician Cathy O'Neil has documented how bail extension rulings by judge became increasingly racist after the judges were advised by an AI tool whose advice was based on data from racist moment in the past that used data from previous decades. Bias riddles contemporary AI products today.[20] Even GPT-3 cannot be trusted to summarize even relatively simple prose without the service of a trained fact checker.[21] The algorithms designed to correct for bias, meanwhile, rarely go to the root of the problem, because most of them were designed by information scientists who lack a familiarity with how the bias got into the data in the first place. Today's AI is simply not capable of performing

[20] Tarleton Gillespie, *The Relevance of Algorithms* (Cambridge, MA: MIT Press, 2014), https://academic.oup.com/mit-press-scholarship-online/book/14976/chapter/169333383; Cathy O'Neil, *Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy* (New York: Crown, 2016); Safiya Umoja Noble, *Algorithms of Oppression: How Search Engines Reinforce Racism* (New York: NYU Press, 2018); Wendy Hui Kyong Chun, *Discriminating Data: Correlation, Neighborhoods, and the New Politics of Recognition* (Cambridge, MA: MIT Press, 2021); Kate Crawford, *The Atlas of AI*.

[21] For my investigations of bias in the historical performance of GPT-3, see "The Dangerous Art of Text Mining," talk at the Alan Turing Institute, London, UK (July 2022), www.youtube.com/watch?v=-V7WZaUk50A. I've applied it to summarizing my own books, and found that there are errors. Where I set out formal arguments, reviewing the historiography of the field and then stating my own position as a refutation of those arguments, GPT-3 is just as likely to summarize the texts that I'm refuting as fact. It has no preference for my own argument or the evidence I've presented; it will summarize anything. It can summarize them with alacrity and questionable accuracy.

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at the rigorous standard of truth that scholars in the liberal arts have long expected of each other.

The cost to society of data science ignoring the liberal arts is enormous. It is represented in quantifiable terms by the lakes of unused textual data that cannot be effectively exploited by today's AI tools. Statisticians estimate that, in 2017, phone users around the world sent 6.8 billion text messages a day.[22] In 2018, the world's population was 86.5 percent literate, and that literate bulk of humanity composed roughly 281 billion emails per day.[23] Indeed, our civilization produces a shocking amount of text every year. Since the advent of electronic systems, such as word processors and webpages, for authoring and sharing text, the amount of text created and preserved in analyzable form has dwarfed earlier numbers. If printed out, the text on the internet itself would consume 136.2 billion pieces of paper.[24] Grand as these figures may be, they are modest compared with the wealth of disposable text that humans generate in the course of continuously communicating, redacting, and recirculating information.[25] One approximation of that number, however, is that the US federal government employees print out about 18 billion pages of documents every year. "Big data just gets bigger and bigger and bigger," wrote Aiden and Michel, reflecting problems of scale from 2013.[26] For want of effective text mining strategies, an ocean of data has gone virtually untouched.

The want of interdisciplinary approaches to text mining that span humanistic questions and quantitative approaches also limits what industry can do with the data it already has. During a conversation with an executive at a streaming movie service, I recently had the occasion to explain the recent successes that define the modern field of text mining

[22] Megan Morreale, "Daily SMS Mobile Usage Statistics," *SMSEagle*, March 6, 2017, www.smseagle.eu/2017/03/06/daily-sms-mobile-statistics.

[23] For literacy, see UNESCO Institute for Statistics, uis.unesco.org. For emails, see J. Clement, "Daily Number of E-Mails Worldwide 2023," *Statista*, August 9, 2019, www.statista.com/statistics/456500/daily-number-of-e-mails-worldwide.

[24] George Harwood and Evangeline Walker, "How Much of the Amazon Would It Take to Print the Internet?," *Journal of Interdisciplinary Science* 4 (2015): 10–11.

[25] Executive Office of the President, Social and Behavioral Sciences Team, *Annual Report* (2015): 46 (with thanks to J. Nathan Matias), https://sbst.gov/download/2015%20SBST%20Annual%20Report.pdf?fbclid=IwAR1_A_UvH6V5zPzCP1FxJzJB4zsKwp8H7bNcoO3qVIgta5V-EgZsIUtRI9c.

[26] For 18 billion pages, see http://opm.gov/policy-data-oversight/data-analysis-documentation/federal-employment-reports/historical-tables/total-government-employment-since-1962. For quotation, see Erez Aiden and Jean-Baptiste Michel, *Uncharted: Big Data as a Lens on Human Culture* (New York: Riverhead Books, 2013): 11. Aiden and Michel refer to big data in general rather than the problem of text.

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in the liberal arts. Curious about my projects, he asked what text mining could do for his own industry and I told him about a dozen new studies from the digital humanities. I explained how scholars in the digital humanities had been counting the underrepresentation of black actors in Hollywood scripts, and observing the structural differences between novels written by women and those written by men.

Because the executive's job was to buy and sell scripts that target the preferences of a wide range of viewers, I offered speculation about how text mining might affect his industry. I imagined that he might do well to hire a digital humanist as these scholars, in recent years, have been modeling the imaginary dimensions of identity and plot, and creating variables that could be measured against the voluminous data the executive had been gathering.

While the company already knew a great deal about which movies were hits and which were duds, a digital humanist could create a model based upon conceptual categories that would explain their viewing behavior and thus offer them movies that best fit their needs. If women stop watching twenty minutes in, for example, near a scene when the men are doing all the talking, a humanist could train an algorithm to recognize the common pattern and its effects on viewing patterns.

The executive agreed. He was thoroughly hooked on the algorithms I described, and he wanted to add them to his business model. But here's the rub, he confessed: he would never be able to sell his bosses on hiring a PhD in literature – a field that has produced some of the finest scholarship on text mining applied to culture. The prejudice against the humanities in Silicon Valley went too deep, he explained.

Creating smart data requires becoming smart *about* data; and if we want the data to provide the most complete reflection of human experience possible, we must try to make a happy marriage between traditional tools of reasoning that have been honed over centuries, and contemporary practices with data.

Having shunned an active collaboration with humanists, Silicon Valley is not nearly as smart as it could be. Big tech needs an engaged, technical conversation about radically improving the way we work with data to model human experience. Many civilizations achieved mass literacy sometime in the nineteenth century; but that didn't mean that every reader could participate in a conversation about hermeneutics. Silicon Valley hires hundreds of computer scientists with basic training in natural language processing, but that doesn't mean they understand what a critical engagement with human experience looks like. If the culture of Silicon Valley added teams of professional humanists to its teams that work on data analysis, this marriage

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could give birth to a data-driven revolution far more subtle, and far more accurate, than either camp could manage on their own.

For such a thing to happen, however, hybrid thinking across data science and the liberal arts must become routine in the places that drive research innovation – in the universities themselves. The $100 million gifts that have created new schools of information at Cornell University, Harvard, MIT, Stanford, and Johns Hopkins have so far made room for very few scholars with a liberal arts background. The exclusion of the humanities from data science centers up to this point is understandable. It is, after all, only in the last few years that a real watershed has been crossed where the contribution of historical thinking to text mining is visible. Dozens if not hundreds of colleagues have contributed to that effort, some of them members of the Temporal Cartography Society who work with text on paper in dusty archives, and some of them former members who ventured into the no man's land and there met up with a few brave data scientists from the world without history. My modest contribution is to assemble the discoveries of others, alongside a few of my own, into a teachable series of problems about how historical questions can reshape data science into a smarter discipline.

About the Chapters

Part I of this book opens with a catalog of dangers about where contemporary data practices are failing. Drawing on a decade of interdisciplinary research, it lays out best practices of how liberal arts approaches (especially those from history) can make work with textual data more robust. Chapter 1 tours a series of high-stakes disasters from data science, where papers retracted after publication have demonstrated how riddled contemporary data-driven practices are with error. It categorizes the major typology of dangers that lead analysts into trouble, drawing especial attention to insufficient engagement with historical context and fantasies of predictive knowledge that cause analysts to rush toward law-like conclusions without stopping to theorize the work algorithms must do to uncover the logic of the past. Chapter 2 proposes a series of remedies for poor thinking with data, especially care about the provenance of data and an interest in the fundamentals of historical thinking can guide data-driven research processes. The chapter introduces several teams of interdisciplinary analysts whose work forms the inspiration for this book. Their ranks include traditional scholars from the liberal arts who collaborate with information

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scientists as well as scholars like myself who have trained in liberal arts subjects but sometimes work with data and code. Chapter 3 dives deeper into the problem of treating words with care, explaining how words can act as barriers to interpretive understanding but also keys that unlock powerful analyses of the past. Chapter 4 extrapolates a theory of a research process capable of working between qualitative questions and quantitative tools. It proposes a series of steps for how a "critical" encounter with a digital archive and quantitative algorithm might proceed in such a way as to preserve humanistic concerns with meaning, bias, silence, and perspective. The result is a "critical search" for the past, which can form part of a larger agenda of *re*-search, but which is fundamentally better suited to humanistic questions than searches conducted with out-of-the-box tools like wordclouds. Chapter 5 steps back to the purposes that data scientists sometimes bring to historical data, investigating one of the chief subjects of misunderstanding and conflict between quantitative and qualitative approaches. It asks head-on whether excavations into the past can support a "predictive" theory of the future. Reviewing historians' positions on this subject reveals both numerous objections to predictive thinking with historical data as well as a handful of important opportunities.

Part II of this book explores how ideas from the philosophy of history can help us to study historical change. Drawing on the scholarship of many contemporary philosophers and historians who have wrestled with the problem of how we know about the past, this Part argues that the past is a multidimensional object, in that long-term forces and short-term shifts, experience and memory, all work together to shape the past, their interactions constantly changing. Chapter 6 draws on the philosophy of history to propose a series of categories of temporal experience appropriate as subjects for data-driven modeling.

Next, we consider question of whether mathematics, data, and modeling can expand the analyst's ability to apprehend the multiple dimensions of change in the historic past. To that purpose, I present a concrete series of case studies where I engage datasets around a theory of historical change. In Chapter 7, I deploy a series of text-mining methods to study how the process of memory changed in nineteenth-century Britain. I present multiple possible quantitative approaches that open up different dimensions of the past, and I show how each approach, and each subinvestigation within each approach, adds nuance and complication to our understanding of the past. Chapter 8 discusses algorithms that can be used to detect the distinctiveness of different eras, and explores how those algorithms can be adjusted to highlight the uniqueness of

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experiences that took place over a day, week, year, or decade. Chapter 9 uses data to study how individuals influence each other. In Chapter 10, I show that algorithms can be used to investigate slow-moving and quick-moving pulses of temporal change at any moment in the past; they can also identify the trends that were at the end of their life, fossils that represented a temporary fad, and other shapes of time that have traditionally been identified through qualitative means. I also theorize what it means to critically adjust an algorithm to a historical question. Chapter 11 introduces theories of "modernization" and suggests several algorithmic strategies for looking at long-term development. At the same time, this chapter proposes important limits to the use of data to revisit the most important questions about modernity, poverty, and marginalization that historians have treated in other ways. In Chapter 12, I investigate a concrete historical question: when and how attacks on environmentalists in the US Congress delayed action on climate change. It argues against the testing of hypotheses in the style of many machine-learning papers, drawing on historical argumentation to propose a process where meaningful insight about the past is the standard of discovery. I also demonstrate, to a degree lacking in the other chapters, what it looks like to work between a data visualization and in-text mentions. Alternating data-driven work with the traditional methods of history makes for a more accurate, nuanced, and robust practice, even when the results are circulated in the form of a data-driven visualization.

In the data-driven case studies of Part II, AI and statistics are a subject for critical inquiry. Of each algorithm, we will ask: Does this approach get us closer to real knowledge, as understood by contemporary historical theories of how memory, temporality, or causality work? How might small adjustments in the parameters of the algorithm or the controlled vocabulary produce different results?

The individual chapters in the Part II will introduce algorithms from vintage statistics such as *divergence*, modern AI approaches to clustering such as *topic modelling,* and machine-learning dependencies such as *word embeddings*. Nonspecialists need not shut the book in alarm; each algorithm will be introduced in accessible language, and the focus of each chapter will remain on the larger question of how hybrid scholarship requires critical thinking about the fit between data, algorithms, and models of historical change.

Some caveats are important. While certain historical "discoveries" are revealed in the middle chapters of this book, I am not primarily concerned in this book with incontrovertible historical argumentation, but

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rather with a practice designed to combine historical insight with algorithmic rigor. Many of the "discoveries" of data science in recent years merely reproduce standard knowledge; to this end, I am clear where my exercises in data science do the same, although my case studies generally aim for a higher standard of insight. I also emphasize where inadequate data means that the data-driven discoveries in question need to be traced back to archives for the historical explanation to be complete. Chapter 12 looks in greater detail into the notion of "explanation" in history, and how it is that data-driven discoveries can rise to the high standard that historians hold for actual discovery. It is an axiom of the method presented here that data-driven analysis is insufficient for many historical problems. A savvy analyst must apply her skills of skepticism to each phenomenon apparent in the data, validating each discovery by work with the text. Only then can data science rise to the ideal of complete explanation to which historians aspire.

Finally, Part III steps back from detailed engagements with methods to ask what can be accomplished when critical thinking is brought to methods. Chapter 13 looks at how data helped historical linguists to purge centuries of racial bias from their map of the relationships between world languages. It takes a long-term view on the development of disciplines to imagine what our age of reform and data might contribute to other foundational systems of understanding in the humanities and social sciences. Chapter 14 investigates the future of research. It articulates a call for philosophers of history to help us to understand more of the dimensions of the past and for data scientists to use historical understanding to investigate the limits and promise of their models. It envisions a robust future for practices of AI and data science in the future, one where (some) humanists and (certain) historians have an important role to play. It calls on us all to contribute to the collective understanding of where text mining can answer big questions and where traditional research is warranted. Finally, it glances at some public uses of text mining for the purpose of transparency and political reform.

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Ultimately, most humanists are already concerned with evidence and analysis. Understanding the past with accuracy and precision is a matter of evidence, and the humanities share this concern for reality with the sciences. Reading the speeches of presidents, comparing novels, studying paintings, or parsing old maps give us evidence, and many humanists pride themselves on their superior talents of attention, comparison, and interpretation.

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Text mining can benefit from humanists' careful discernment about the dangers of biased data and the promise of careful categorization of experience. Like other humanistic enterprises, text mining is therefore best pursued with some knowledge of historical context, of the limits of each body of evidence, and the methodological questions suited to those objects in particular.

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